



GS FOUNDATION 2024-25

GEOGRAPHY - 16

Minerals and Rocks

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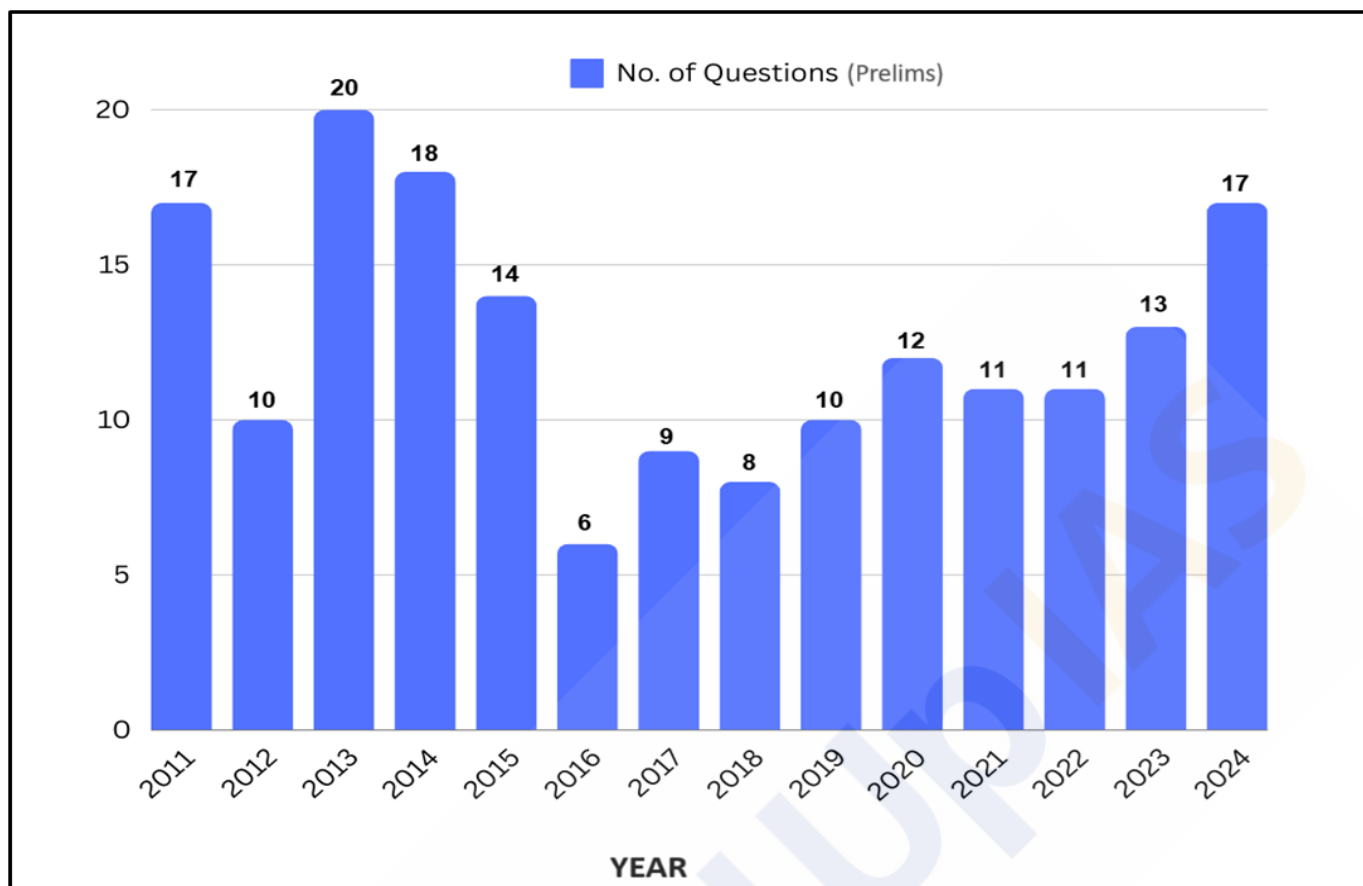
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Minerals

Introduction:

The Earth is composed of various elements, existing in solid form on the outer layer and in hot, molten form in the interior. **About 98% of the Earth's crust is made up of eight primary elements - oxygen, silicon, aluminium, iron, calcium, sodium, potassium, and magnesium.** The remaining 2% consists of elements such as titanium, hydrogen, phosphorus, manganese, sulphur, carbon, nickel, and others.

The elements in the earth's crust are rarely found exclusively but are usually **combined with other elements to make various substances.** Though the lithosphere contains a limited number of elements, these elements combine in numerous ways, resulting in a wide variety of minerals. **Over 2,000 minerals have been identified in the Earth's crust,** but the **majority belong to six major rock-forming mineral groups.**

The primary source of all minerals is the hot magma within the Earth. As magma cools, mineral crystals form in a specific sequence, solidifying to create rocks. Minerals such as coal, petroleum, and natural gas are organic and found in solid, liquid, and gaseous forms, respectively.

Mineral

A mineral is a naturally occurring substance that can be either organic or inorganic. It possesses an orderly atomic structure, along with a specific chemical composition and distinct physical properties. Typically, a mineral consists of two or more elements. However, there are instances where minerals consist of single elements, such as sulphur, copper, silver, gold, and graphite.

Physical Characteristics of Minerals:

External Crystal Form:

- The external crystal form of a mineral is **determined by its internal molecular arrangement.** This can result in various geometric shapes such as **cubes, octahedrons, and hexagonal prisms.**

Cleavage:

- Cleavage refers to **the tendency of minerals to break along specific planes,** resulting in relatively flat surfaces. This is also a consequence of the **internal arrangement of molecules.** Minerals may cleave in one or more directions and at any angle relative to each other.

Fracture:

- When the internal **molecular arrangement of a mineral is too complex** to form planes of cleavage, it **breaks in an irregular manner.** This type of breakage is known as fracture.

Lustre:

- Lustre describes the **appearance of a mineral's surface without considering its colour.** Each mineral has a distinctive lustre, such as metallic, silky, or glossy.

Colour:

- The colour of a mineral can be a characteristic feature determined by its molecular structure. For instance, malachite, azurite, and chalcopyrite have characteristic colours. Additionally, some minerals may be coloured by **impurities,** as seen with quartz, which can appear white, green, red, or yellow depending on the impurities present.

Streak:

- Streak refers to the **colour of the powdered form of a mineral**. This may be the same as the mineral's apparent colour or different. For example, malachite has a green streak, while fluorite, despite being purple or green, gives a white streak.

Transparency

- Transparency is classified into **three categories** - **transparent**, where light passes through the mineral allowing objects to be seen clearly; **translucent**, where light passes but is diffused, making objects appear blurred; and **opaque**, where no light passes through.

Structure:

- The structure of a mineral pertains to the **specific arrangement of its individual crystals**. This can be fine, medium, or coarse-grained. Additionally, some minerals may exhibit fibrous structures that can be separable, divergent, or radiating.

Hardness:

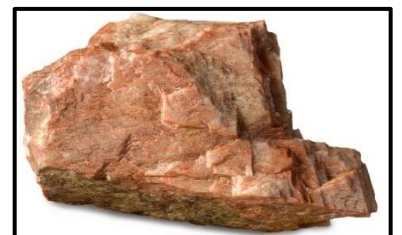
- Hardness measures a **mineral's resistance to being scratched**. This property is quantified on a **scale from 1 to 10** using a reference list of minerals: 1. Talc, 2. Gypsum, 3. Calcite, 4. Fluorite, 5. Apatite, 6. Feldspar, 7. Quartz, 8. Topaz, 9. Corundum, and 10. Diamond. For context, a fingernail has a hardness of 2.5, while glass or a knife blade is about 5.5.

Specific Gravity

- Specific gravity is the **ratio of the weight of a mineral to the weight of an equal volume of water**. This property is determined by weighing the mineral in air and then in water, and dividing the weight in air by the difference between the two weights.

Major Minerals and Their Characteristics:**Feldspar:**

- Feldspar is **the most abundant mineral on Earth**, constituting nearly **half of the Earth's crust**. It is composed primarily of **silicon, oxygen**, and varying amounts of **sodium, potassium, calcium**, and **aluminium**. Feldspar exhibits a **light cream to salmon pink colour** and is widely used in ceramics and glass manufacturing.

**Quartz:**

- Quartz is another essential mineral found in **abundance in rocks like sand and granite**. It is composed of **silica** and is characterized by its hardness and **insolubility in water**. Quartz is typically **colourless or white** and has diverse applications, including in the radio and radar industries.

**Pyroxene:**

- Pyroxene is a common mineral containing **calcium, aluminium, magnesium, iron**, and **silica**. It accounts for approximately **10% of the Earth's crust** and is even found in **meteorites**. Pyroxene minerals often exhibit **green or black coloration**.



Amphibole:

- Amphiboles are a group of minerals containing **aluminium, calcium, silica, iron,** and **magnesium**. They constitute about **7% of the Earth's crust** and are commonly found in **green or black varieties**. **Asbestos**, a commercially important material, is derived from amphiboles. **Hornblende** is a well-known example of an amphibole mineral.

**Mica:**

- Mica is a mineral composed of **potassium, aluminium, magnesium, iron,** and **silica**. It forms approximately **4% of the Earth's crust** and is commonly found in igneous and metamorphic rocks. Mica's unique properties make it valuable in the electrical industry.

**Olivine:**

- Olivine is a mineral rich in **magnesium, iron,** and **silica**. Often found in **greenish crystals**, it is commonly associated with basaltic rocks. While not as abundant as some other minerals, olivine has applications in the **jewellery industry**.

**Other Important Minerals:**

- Beyond these primary minerals, several other minerals like **chlorite, calcite, magnetite, hematite, bauxite,** and **barite** are present in varying quantities within rocks. These minerals contribute to the overall composition and properties of the rock formations.

Classification of Minerals:

Minerals can be broadly classified into **two categories - metallic and non-metallic minerals**. This classification is based on the presence or absence of metal content in the minerals.

Metallic Minerals:

Metallic minerals **contain metal content** and are further **subdivided into three types** based on their specific properties and uses:

- Precious Metals** - These are rare and **valuable metals**, including **gold, silver,** and **platinum**. They are often used in jewellery, electronics, and as investment assets.
- Ferrous Metals** - This group includes **iron and other metals that are typically mixed with iron** to create various kinds of steel. These metals are crucial in the **construction and manufacturing industries** due to their strength and durability.
- Non-Ferrous Metals** - These metals **do not contain iron** such as **copper, lead, zinc, tin,** and **aluminium**. They are used in a wide range of applications, from electrical wiring (copper) to packaging (aluminium) and batteries (lead).

Non-Metallic Minerals:

Non-metallic minerals **do not contain metal content** and have a diverse range of applications. Examples include:

- Sulphur** - Used in the production of **sulfuric acid**, which is a key industrial chemical.
- Phosphates** - Essential for the manufacture of fertilizers, which are critical for agricultural productivity.
- Nitrates** - Also used in **fertilizers** as well as in **explosives**.

Additionally, **cement is a notable product made from a mixture of non-metallic minerals**. It is a fundamental material in the construction industry, used for making concrete and other building structures.

Important Metallic Minerals:

Iron Ore:

Iron ore is a term used to describe rocks and minerals from which metallic iron can be economically extracted. These **ores are rich in iron oxides** and are typically found in various colours such as **dark grey, bright yellow, rusty red, and deep purple**. Iron ores are classified based on the percentage of metallic iron present:

- **Magnetite** - The best quality iron ore, containing more than 70% metallic iron. It is black and exhibits magnetic properties, making it highly valuable.
- **Hematite** - This ore contains about 65-70% metallic iron and is usually red or brown in colour. It is one of the primary ores used for iron extraction.
- **Siderite** - Siderite has a lower iron content, about 40-50%, and contains more impurities. It is brown and consists of a mixture of iron and carbon.
- **Limonite** - A low-grade iron ore with below 50% metallic iron content. It is mostly yellow and contains lime, making it self-fluxing.

World Distribution of Iron Ore and Alloy Metals:

Alloy metals are essential components in the production of high-quality steel. These elements, which include **manganese, molybdenum, nickel, chromium, vanadium, tungsten, and cobalt**, enhance the properties of steel, making it suitable for various industrial applications. **Steels containing at least 1.6% manganese** or any other non-ferrous metal are known as **alloy steels**.

Alloy Metals

A metal alloy is a substance that combines more than one metal or mixes a metal with other non-metallic elements.

Ferrous Metals:

Manganese:

- Manganese is **crucial for producing good-quality steel**, as it decreases brittleness and imparts strength. India possesses significant high-grade manganese ore deposits in **Madhya Pradesh, Odisha, Maharashtra, Goa, Jharkhand, Tamil Nadu, and Karnataka**. The country exports manganese to the USA, the UK, France, and Japan.
 - Globally, **South Africa accounts for about 78% of the world's identified manganese resources**.

Cobalt:

- Cobalt is primarily sourced from **Congo (Kinshasa)**, which **supplies more than half of the world's cobalt production**. This metal is vital for various high-tech and industrial applications.

Molybdenum:

- Molybdenum is extensively used in high-temperature alloys for components in gas **turbines, jet engines, and turbo-superchargers**. It does not occur in nature in a free state, with molybdenite being its principal ore. In India, molybdenum is intermittently produced as a **by-product from the uranium ore of Jadugoda mine in Jharkhand**.
 - Globally, **about two-thirds of molybdenum production is a by-product of copper mining**, with leading producers including **China, Chile, the USA, Peru, and Mexico**.

Tungsten:

- A significant amount of tungsten is recovered through **recycling scrap** tungsten products. India's only operative mines, Degana in Rajasthan and Chendapathar in West Bengal, are now closed.
 - Major global producers of tungsten include **China, Vietnam, Russia, the UK, and Bolivia**.

Vanadium:

- Vanadium's leading producers and reserves are found in **China, Russia, South Africa, and Brazil**. India meets its vanadium demand mainly through imports and some from red mud sludge.

Chromium:

- In India, chromium deposits are found in **Bihar, Karnataka, Maharashtra, Odisha, and Tamil Nadu**.
 - The leading global producers of chromium are **South Africa, Kazakhstan, and India**.

Nickel:

- In India, **Odisha** is the leading producer of nickel, followed by **Karnataka, Kerala, and Rajasthan**.
 - Globally, the leading producers are Indonesia, the **Philippines, New Caledonia, Canada, and Australia**, with **maximum reserves in Australia**, followed by Brazil, Russia, and Cuba.

Non-Ferrous Metals:**Aluminium:**

- Aluminium is a **lightweight and durable metal** primarily **obtained from bauxite**. In India, Odisha is the leading producer, accounting for half of the nation's bauxite production, followed by **Gujarat, Jharkhand, Maharashtra, Chhattisgarh, Tamil Nadu, and Madhya Pradesh**.
 - Globally, **Australia, China, Brazil, Indonesia, Guinea**, and India are the chief producers of bauxite.

Copper:

- Copper occurs naturally and is often **found mixed with gold, silver, lead, and zinc**. Major copper mines in India include the **Khetri** copper belt in **Rajasthan**, the **Singhbhum** copper belt in **Jharkhand**, and the **Malanjkhand** copper belt in **Madhya Pradesh**.
 - Leading copper-producing countries are **Chile, Peru, China, and the USA**.
- **Copper Alloys:**
 - **Brass** - An alloy of copper and zinc.
 - **Bronze** - Typically refers to copper-tin alloys but can include other copper alloys like aluminium bronze.
 - **Carat Silver and Gold Alloys** - Copper is a vital component in these alloys, used in the jewellery industry to modify colour, hardness, and melting point.
 - **Cupronickel** - An alloy of copper and nickel used in low-denomination coins.

Lead:

- In India, the **Zawar mines in Udaipur district of Rajasthan** are the major producers of lead ore.
 - Globally, the leading producers are **China, Australia, the USA, and Peru**, with **Australia holding the maximum reserves**.

Zinc:

- India's zinc production mainly comes from the **Zawar mines in Rajasthan** and the **Arcot districts of Tamil Nadu**.
 - Principal zinc-producing countries are **China, Peru, India, and Australia**, with **Australia having the largest reserves**.

Tin:

- The principal ore of tin is **cassiterite**. In India, **Chhattisgarh** is the only tin-producing state.
 - Globally, **China** has the maximum reserves, followed by **Indonesia, Brazil, Australia, and Bolivia**.

Mica:

- In India, **Gaya-Hazaribagh belt in Jharkhand** accounts for **60% of production**. Nellore in Andhra Pradesh and Rajasthan are other areas of production.
 - Large deposits of mica-bearing rocks are known to exist in countries such as **Brazil, India, and Madagascar**.

Precious Metals:**Silver:**

- In India, the majority of silver production originates from the **Zawar mines of lead in Rajasthan**. Additionally, silver is extracted along with gold from the **Kolar gold fields** and with copper from the **Manbhum region** (Dhanbad and Purulia).
 - Globally, **Peru** holds the maximum reserves of silver, making it a significant player in the silver market.

Platinum

- India's platinum deposits are found primarily in **Orissa** and **Hanumalpur in Karnataka**.
 - **South Africa** stands as the largest producer of platinum worldwide, followed by **Russia, Zimbabwe, Canada, and the USA**. These countries contribute substantially to the global supply of this precious metal.

Gold

- Gold in India is predominantly found in a relatively **pure state**, manifesting as thin veins within rocks. The country has three major goldfields located at **Kolar and Hutti in Karnataka** and **Ramgiri in Andhra Pradesh**.
 - On the global stage, **China, Australia, Russia, and the USA** are leading producers of gold, significantly influencing the international gold market.

Precious Stones:**Diamonds**

- Historically, India was the sole source of diamonds until 1726. Today, major diamond production in India is concentrated in the **Panna mines of Madhya Pradesh**.
 - Internationally, **Russia holds the maximum reserves** of diamonds, followed by **Congo, Australia, and Botswana**, positioning these countries as key players in the diamond industry.

Blood Diamonds

*Blood diamonds, also known as **conflict diamonds**, are mined in areas controlled by rebel forces opposing the government. **These diamonds fund armed conflicts and violent campaigns**. During the 1990s, conflict diamonds were prevalent in **African nations like Congo, Sierra Leone, and Angola**.*

*To combat this, the **Kimberley Process Certification Scheme** was established in **2003** to verify that diamond exports are free from conflict diamonds. India served as the chair and host of the Kimberley Process in 2019.*

Important Non-Metallic Minerals:

Asbestos:

- Asbestos is a group of **naturally occurring fibrous minerals** used for their **heat-resistant properties**. In India, significant deposits of asbestos are found in **Jharkhand, Odisha, Madhya Pradesh, Karnataka, and Andhra Pradesh**. These regions contribute to the country's asbestos production.
 - Globally, the leading producers of asbestos include **Russia, Kazakhstan, China, and Brazil**. The mineral is utilized in various industries due to its durability and resistance to heat and chemicals.

Sulphur:

- Sulphur is predominantly **found in igneous rocks** and is essential for various industrial processes. Approximately **75% of the sulphur mined is used in the production of sulphuric acid**, a crucial chemical in fertilizer manufacture. Additional uses of sulphur include its application in the production of fertilizers and other chemical compounds. In India, major sulphur-producing states include **Chhattisgarh**.
 - On the global stage, **China** is the dominant producer of sulphur, with other significant producers being the **USA, Russia, Canada, and the UAE**.

Gypsum:

- Gypsum is a versatile mineral used primarily in the **construction industry**, particularly in the manufacture of plaster and wallboard. In India, **Rajasthan** holds the majority of gypsum reserves, with additional deposits located in **Jammu and Kashmir, Tamil Nadu**, and other regions.
 - **China** is the leading global producer of gypsum, followed by the **USA, Iran, Thailand, Turkey, and Italy**. The mineral's extensive use in construction makes it a critical resource for building and infrastructure development.

Salt:

- Salt is obtained from various sources, including **seawater, lake water, subsoil water, and rock salt beds**. Rock salt, derived from solid salt beds, is commonly used in the chemical industry and as commercial table salt. In India, **Gujarat is the largest producer of salt**, utilizing methods such as evaporation from seawater and lake water.
 - On a global scale, **China** is the leading producer of salt, followed by **the USA, India, Germany, and Canada**. The mineral's wide range of uses, from culinary to industrial applications, underscores its significance in daily life and industry.

Rocks

Introduction:

The Earth's crust, the outermost layer of the lithosphere, is crucial as it is where human activity occurs. Composed primarily of rocks, the crust varies greatly in hardness, colour, and composition. Rocks are aggregates of minerals, which themselves are made up of two or more elements in a specific ratio.

Types of Rocks - Rocks are categorized based on their formation processes and appearance into **three main types**:

- Igneous Rocks
- Sedimentary Rocks
- Metamorphic Rocks

Igneous Rocks:

The word igneous is derived from the Latin word '**ignis**' meaning **fire**. Igneous rocks, also known as **primary rocks**. They are formed from the **solidification of magma** (molten rock below the Earth's surface) **and lava** (molten rock that reaches the surface). As the first rocks on Earth, igneous rocks originate under conditions of high temperatures and are characterized by the **absence of fossils**. **Approximately 95% of the outermost 16 kilometers of Earth's volume consists of igneous rocks**. They are typically hard, massive, and crystalline due to their magmatic origin.

Classification of Igneous Rocks:

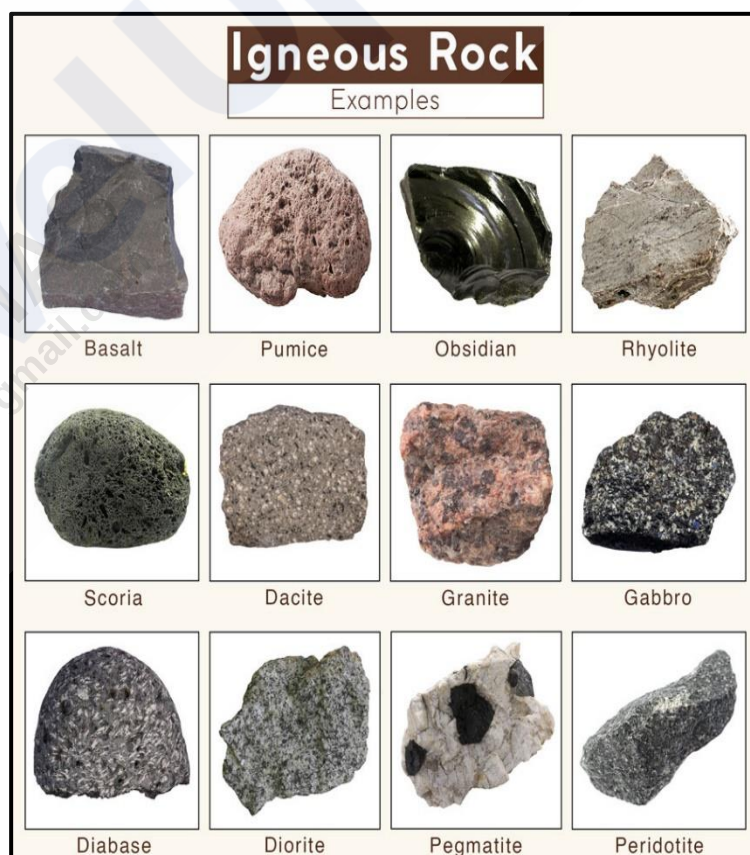
Based on Cooling and Origin:

• Intrusive Igneous Rocks (Plutonic Rocks):

- These rocks form from **magma that cools slowly at great depths** within the Earth. The slow cooling process allows the formation of large mineral grains. Plutonic rocks are **usually medium to large grained** due to their **slow cooling rates**. They appear on the surface only after significant geological processes, such as uplift and denudation, expose them.
- **Examples - Granite, Diorite, and Gabbro** are common plutonic rocks. They are typically exposed at the surface through processes of denudation and erosion.

• Extrusive Igneous Rocks (Volcanic Rocks):

- These rocks form **from lava that cools quickly either just below the surface or at the surface**. Due to their **rapid cooling, volcanic rocks are often fine-grained and can be smooth**. They are typically lighter in colour and less dense compared to plutonic rocks.
- **Examples - Basalt** is a well-known **volcanic rock**, with significant examples including the **Deccan Traps in India** and the **Snake Plateau in Colombia**.



- **Intermediate Rocks (Hypabyssal or Dyke Rocks):**

- These rocks have cooling conditions and characteristics that **fall between those of plutonic and volcanic rocks**. Intermediate rocks display a range of textures and mineral compositions that bridge the gap between intrusive and extrusive rocks.
- **Example - Dyke** rocks are semi-crystalline in structure.

Based on Composition:

- **Acidic Rocks:**

- Acidic rocks are characterized by their **high silica content**, which can reach up to **80%**. This composition primarily includes **quartz and feldspar**. The remaining components consist of aluminium, alkalis, magnesium, iron oxide, and lime.
- Due to **the lower presence of heavier minerals** such as iron and magnesium, acidic rocks are **less dense and lighter in colour** compared to basic rocks.
- These rocks constitute the **sial portion of the Earth's crust**. The high silica content in acidic magma causes it to **cool rapidly, resulting in limited flow and spread**. Consequently, high mountain ranges are often formed from these rocks. Acidic rocks are known for their **hardness, compactness, and resistance to weathering**.
- **Example - Granite, quartz, and feldspar.**

- **Basic Rocks:**

- Basic rocks have a **lower silica content**, around **40% or less**, with a **significant amount of magnesia, up to 40 percent**. The remaining composition includes **iron oxide, lime, aluminium, alkalis, and potassium**.
- Due to their low silica content, the parent material of basic rocks **cools more slowly and flows more extensively, leading to the formation of plateaus**.
- The presence of heavier elements gives these rocks a **dark colour**. Basic rocks are generally **less hard and more susceptible to weathering**.
- **Example - Basalt, gabbro, and dolerite.**

Economic Significance of Igneous Rocks:

- Since magma is the chief source of metal ores, many of them are associated with igneous rocks.
- The minerals of great economic value found in igneous rocks are magnetic iron, nickel, copper, lead, zinc, chromite, manganese, gold, diamond and platinum.
- Amygdales are almond-shaped bubbles formed in basalt due to escape of gases and are filled with minerals.
- The old rocks of the great Indian peninsula are rich in these crystallised minerals or metals.
- Many igneous rocks like granite are used as building materials as they come in beautiful shades.

Sedimentary Rocks:

Sedimentary rocks are formed through the **successive deposition of sediments**, which are fragments derived from previously existing rocks such as igneous, metamorphic, or older sedimentary rocks. These rocks typically exhibit a **layered or stratified structure**, with strata varying in thickness from a few millimeters to several meters, earning them the alternative name of stratified rocks. **Often, fossils—solid parts or impressions of prehistoric plants or animals—are found within these strata.**

Sedimentary rocks cover about **75% of the Earth's crust**, yet they occupy only about **5% of its volume**, primarily because they are found in the upper layers of the crust.

Formation and Classification:

Sedimentary rocks originate from sediments produced by the denudation of various rock types, which includes weathering and erosion. These rocks can be classified based on their mode of formation:

- **Mechanically Formed Sedimentary Rocks:** These rocks are created by mechanical agents such as **running water, wind, ocean currents, and ice**. Examples include:
 - **Arenaceous Rocks** - These rocks have a **high sand content and larger particles**, making them **hard and porous**. They are **excellent reservoirs for groundwater and petroleum**. **Sandstone** is a typical example.
 - **Argillaceous Rocks** - Comprising more clay and finer particles, these rocks are softer and generally less porous. Examples include claystone and shale.
- **Chemically Formed Sedimentary Rocks:** Formed from the **evaporation of water containing minerals**, these rocks can include:
 - **Limestone, Halite, Potash** - These rocks result from chemical precipitation, often observed in environments like **salt lakes and springs**.
- **Organically Formed Sedimentary Rocks:** These rocks are formed from the **remains of plants and animals** that undergo changes due to heat and pressure over time. Examples include:
 - **Coal and Limestone:** Depending on their composition, they can be classified as **either calcareous** (e.g., limestone, chalk) **or carbonaceous** (e.g., coal).

Chief Characteristics:

- **Stratification** - They are layered, consisting of multiple strata.
- **Informative Geological Records** - These rocks preserve detailed geological records due to various physical and biological processes such as weather patterns and fossil formation.
- **Fossiliferous Nature** - They often contain fossils of plants and animals.



- **Porosity** - Generally, they are porous, allowing water to percolate through them.

Spread of Sedimentary Rocks in India:

- **Alluvial Deposits** - Found in the **Indo-Gangetic Plain** and **coastal plains**, these deposits include loam and clay.
- **Sandstones** - Various types of sandstone are present in Madhya Pradesh, eastern Rajasthan, parts of the Himalayas, Andhra Pradesh, Bihar, and Orissa.
- **Vindhyan Highland** - This region in central India is known for its sandstones, shales, and limestones.
- **Coal Deposits** - Significant coal deposits are found in the river basins of the Damodar, Mahanadi, and Godavari, associated with Gondwana sedimentary deposits.

Economic Significance of Sedimentary Rocks:

- Sedimentary rocks are not as rich in minerals of economic value as the igneous rocks.
- But important minerals such as hematite iron ore, phosphates, building stones, coals, petroleum and material used in the cement industry are found.
- The decay of tiny marine organisms yields petroleum. Petroleum occurs in suitable structures only.
- Important minerals like bauxite, manganese, tin, are derived from other rocks but are found in gravels and sands carried by water.
- Sedimentary rocks also yield some of the richest soils.

Metamorphic Rocks:

Metamorphic rocks are formed through the process of **metamorphism**, which involves the **recrystallization and reorganization of minerals within a rock due to changes in pressure, temperature, and volume**. This process occurs when rocks are subjected to significant pressure, either through tectonic movements that force rocks to lower levels, or when molten magma coming through the crust interacts with surrounding crustal rocks.

The term "**metamorphic**" itself means "**change of form**," reflecting the transformation that these rocks undergo during metamorphism. Most rocks in **mountainous regions** show evidence of change.



Characteristics of Metamorphic Rocks:

- During metamorphism, the minerals within the rock may become **aligned in layers or lines**, a feature known as **foliation or lineation**.
- In some cases, minerals or materials of different types are arranged into alternating layers of varying thickness, which is referred to as **banding**.
- Common examples of metamorphic rocks include **gneissoid, slate, schist, marble, and quartzite**.

Causes of Metamorphism:

- **Orogenic (Mountain Building) Movements** - Orogenic movements, associated with **mountain building**, involve **folding, warping, and high temperatures** that alter existing rocks. These processes lead to the development of new mineral structures and appearances in the rocks.
- **Lava Inflow** - The intrusion of **molten magmatic material into the earth's crust can subject surrounding rocks to intense temperatures and pressures, causing metamorphic changes**. For instance, the peak of **Mount Everest**, composed of metamorphosed **limestone**, illustrates the effects of lava inflow.
- **Geodynamic Forces** - Geodynamic forces, such as plate tectonics, play a crucial role in metamorphism. The **movement and interaction of tectonic plates** contribute to the conditions necessary for metamorphism.

Types of Metamorphism:

- **Thermal Metamorphism:**
 - Thermal metamorphism occurs when rocks experience recrystallization **under high temperatures**. This type of metamorphism is typically associated with magmatic intrusions.
 - **Example - Sandstone can transform into quartzite, and limestone can become marble** as a result of thermal metamorphism. A notable example is the metamorphosed limestone found in the peak of Mount Everest.
- **Dynamic Metamorphism:**
 - Dynamic metamorphism involves the formation of metamorphic rocks **under high pressure**. This can occur with or without accompanying high temperatures and the influence of chemically charged water.
 - When directed pressure and heat are combined, it leads to significant recrystallization and the development of new rock structures. This is known as **dynamo-thermal metamorphism**.
 - **Example - Under high pressure, granite can transform into gneiss, and clay and shale can change into schist.**

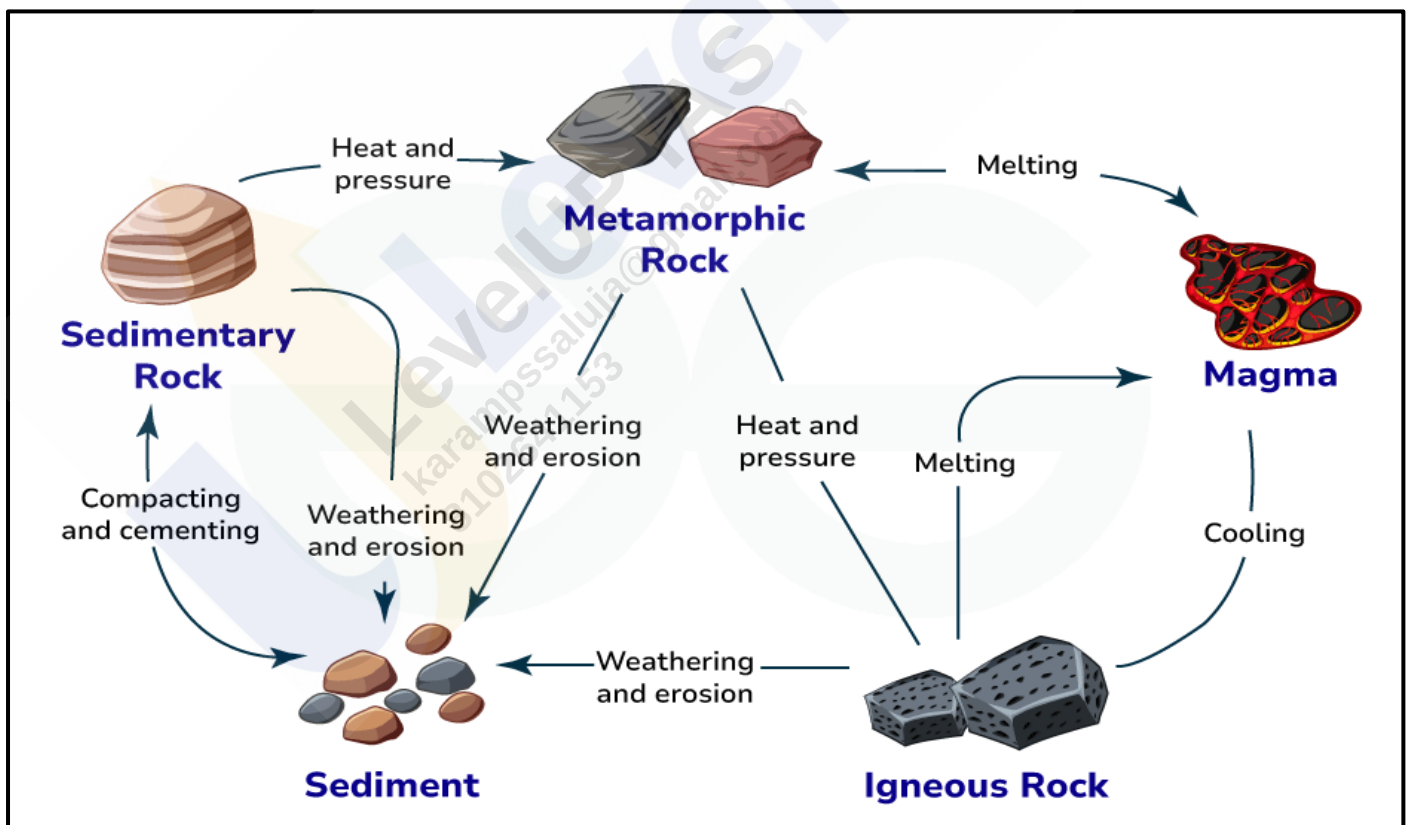
Metamorphic Rocks in India:

- **Gneisses and Schists** - These are prevalent in the Himalayas and extend through Assam, West Bengal, Bihar, Orissa, Madhya Pradesh, and Rajasthan.
- **Quartzite** - This hard rock is commonly found in Rajasthan, Bihar, Madhya Pradesh, Tamil Nadu, and the regions surrounding Delhi.
- **Marble** - Marble deposits are located near Alwar, Ajmer, Jaipur, and Jodhpur in Rajasthan, as well as in parts of the Narmada Valley in Madhya Pradesh.
- **Slate** - Used for roofing and educational writing, slate is found in Rewari (Haryana), Kangra (Himachal Pradesh), and parts of Bihar.
- **Graphite** - This material is present in Orissa and Andhra Pradesh

Rock Cycle:

The rock cycle is a dynamic and **continuous process through which existing rocks are transformed into new types**. This cycle involves the formation, alteration, and recycling of rocks through various geological processes.

- **Igneous Rocks:** These are the primary rocks formed from the cooling and solidification of molten magma. They serve as the **starting point for the rock cycle**.
- **Transformation into Other Rocks:** Igneous rocks can **undergo changes to become sedimentary or metamorphic rocks**. This transformation occurs through **different geological processes**:
 - **Sedimentary Rocks** - Formed from the **accumulation and compaction of fragments derived from the weathering and erosion** of igneous and metamorphic rocks. Over time, these fragments are deposited in **layers and cemented together**.
 - **Metamorphic Rocks** - Created when igneous or sedimentary rocks are **subjected to high pressure and temperature conditions**, causing them to recrystallize and change in mineral composition and structure.
- **Recycling into Magma:** Both sedimentary and metamorphic rocks can be subjected to **high temperatures and pressures**, causing them to melt and **become molten magma**. This molten material then **rises to the surface and cools to form new igneous rocks, completing the cycle**.
- **Subduction Process:** In the Earth's crust, rocks are carried down into the mantle through the process of subduction. As these rocks descend, they **melt and transform into magma**. This magma eventually returns to the surface, contributing to the formation of new igneous rocks.



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